

TRAPNET: Edge-AI Driven Physical Intrusion Response System with Mobile Edge Processing Node and Radar-Vision Fusion

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Abstract— This paper presents TRAPNET, an autonomous Edge-AI physical intrusion containment and emergency response system engineered to overcome the critical limitations of conventional passive Closed-Circuit Television (CCTV) surveillance. Standard security infrastructure relies exclusively on passive recording or delayed remote notifications, leaving perimeter breaches uncontained for 10–15 minutes. TRAPNET executes real-time physical containment within <235 ms of intrusion verification by replacing dedicated microcomputers with a Mobile Edge Processing Node (MEPN)—a high-performance smartphone running quantized You Only Look Once version 8 (YOLOv8) computer vision inference—fused with a 24 GHz Frequency-Modulated Continuous Wave (FMCW) millimeter Wave (mmWave) radar (HLK-LD2410C) for micro-motion breathing signature detection. To guarantee zero-trust operational resilience against camera tampering (such as spray painting, lens occlusion, or physical blinding), an automated OpenCV heartbeat algorithm running on the MEPN monitors frame brightness and structural similarity, instantly delegating primary detection to the mmWave radar without missing an intrusion event. Upon threat confirmation, an electro-mechanical response layer triggers a 12V 60kg solenoid door deadbolt to physically trap the intruder. Concurrently, a multi-layer offline emergency alert engine broadcasts localized Wi-Fi beacon SSIDs, Bluetooth Low Energy (BLE) Eddystone-URL frames, ESP-NOW peer-to-peer mesh alerts across surrounding ESP32 nodes, and long-range GSM SMS notifications with NEO-6M GPS coordinates. Field evaluation demonstrates a 97.6% overall intrusion detection accuracy, zero false lockdown activations under high-energy ambient noise, and full operational capability during complete power and internet grid cuts, realized at a total project cost of ₹5,944.

Keywords— Physical Intrusion Response, Mobile Edge Processing Node (MEPN), mmWave Radar, Sensor Fusion, Edge AI, YOLOv8, ESP-NOW Mesh, Solenoid Lockdown, Anti-Tamper Surveillance.

In contemporary physical asset protection and commercial facility monitoring, security infrastructure remains heavily reliant on passive surveillance architectures such as closed-circuit television (CCTV) cameras, passive infrared (PIR) motion detectors, and remote alarm buzzers. Although modern smart cameras incorporate cloud-based deep learning algorithms for human detection and loitering recognition, traditional setups suffer from a fundamental structural vulnerability: passive recording without real-time physical intervention. In high-value retail burglaries, bank vault breaches, and jewellery store break-ins, the window between initial perimeter compromise and perpetrator escape routinely spans 90 to 180 seconds. Conversely, physical security guard dispatch or police arrival requires an average of 10 to 15 minutes. As a result, standard CCTV configurations act exclusively as post-incident forensic tools rather than active threat containment mechanisms.

To overcome these fundamental limitations, recent pioneering studies have explored multi-modal sensing and edge intelligence. Raimondi et al. [1] proposed mmDetect, utilizing 60 GHz mmWave radar data cubes processed by deep neural networks to achieve resilient human target detection under complex indoor clutter. Building on multi-modal integration, Sengupta et al. [2] introduced a decision-level radar-camera sensor fusion framework leveraging tri-Kalman filtering to maintain target tracking continuity even under severe optical occlusions. Further refining high-precision spatial awareness, Shin et al. [3] fused 3D mmWave radar point clouds with computer vision skeleton estimation, establishing the superior capability of radar-vision synergy for non-intrusive human presence monitoring. Inspired by these foundational advances [1]–[3], physical security systems require active edge-driven actuation to prevent perimeter breaches in real time.

I. INTRODUCTION

Furthermore, visual surveillance systems demonstrate extreme susceptibility to intentional physical tampering and environmental degradation. Intruders frequently neutralize camera feeds using aerosol spray paint, fabric lens covers, power cable severing, or high-intensity infrared laser blinding. Passive Infrared (PIR) motion sensors, while immune to optical spray attacks, are severely limited by their inability to detect stationary intruders who freeze in place, while suffering from high false-alarm rates triggered by thermal convective currents, small domestic animals, or HVAC airflow.

To eliminate these critical operational bottlenecks, this paper introduces TRAPNET (Trapping Network)—an autonomous, multi-sensor, physical intrusion response system operating entirely on decentralized edge hardware. Unlike conventional systems that deploy dedicated single-board microcomputers, TRAPNET leverages a Mobile Edge Processing Node (MEPN)—a high-performance smartphone running Edge-AI deep neural network models—as its primary visual compute engine, coupled with an ESP32 microcontroller hub and a 24 GHz mmWave radar. TRAPNET bridges the gap between threat detection and physical containment by integrating four key technical innovations:

- 1) Mobile Edge Processing Node (MEPN) Vision AI: Utilizes a COTS mobile smartphone running TensorFlow Lite / YOLOv8 Nano computer vision inference to perform real-time human detection and zero-trust anti-tamper optical monitoring directly on device hardware.
- 2) Dual-Mode Radar-Vision Fusion: Fuses 24 GHz FMCW mmWave radar micro-Doppler chest-displacement sensing (detecting motionless human breathing down to 0.5 mm displacement) with MEPN object recognition to achieve near-zero false positive containment decisions.
- 3) Active Physical Containment Layer: Replaces passive alarms with immediate electro-mechanical containment consisting of a 12V high-torque solenoid door deadbolt that physically locks perimeter exits within 15 ms of confirmation, preventing escape prior to responder arrival.
- 4) Quad-Layer Infrastructureless SOS Mesh: Solves internet and cellular jamming vulnerabilities by deploying a local emergency alert engine capable of simultaneous Wi-Fi SSID beacon flooding, BLE Eddystone advertising, ESP-NOW mesh relaying across neighboring ESP32 nodes, and GSM SMS dispatch with NEO-6M GPS coordinates.

II. LITERATURE REVIEW

This section synthesizes 23 published studies across four technical domains that collectively expose the operational gaps in conventional surveillance infrastructure. References follow the bibliography order.

A. Millimeter-Wave Radar & RF-based Human Sensing

Shahbazian and Trubitsyna [6] surveyed WiFi CSI, BLE, and mmWave radar for occupancy sensing and concluded that 24 GHz and 60 GHz mmWave radar offers the best trade-off between privacy, light independence, and spatial resolution. Shrestha et al. [7] trained a Bi-LSTM network on FMCW micro-Doppler spectrograms and classified six human activities with over 90 percent accuracy; importantly, micro-Doppler signatures persist even when a person stands still because chest wall displacement from breathing produces a periodic frequency shift. Shen et al. [8] detected stationary humans at greater than 97 percent sensitivity using range-profile standard deviation analysis on a weak-ranging FMCW radar. Wang et al. [9] applied maximum ratio combining with FastICA on a MIMO FMCW array to separate respiration and heartbeat waveforms at distances up to 2 meters, while Vignoli et al. [10] validated room-scale contactless breathing monitoring using multiple 24 GHz SIMO radar units with Bayesian fusion. Khan et al. [11] implemented a temporal CNN on a System-on-Chip edge device for vital sign extraction at 12.6 ms inference latency. Deng and Zhang [12] achieved the highest precision in this category with a mean heartbeat error of 3.29 percent by combining range-bin FFT processing with an impulse train model, resolving chest displacements below one centimeter.

B. Vision-Based Surveillance & Edge Computing

Billah et al. [4] deployed YOLOv8 across a multi-camera CCTV network for anomalous event detection and reached 96.0 percent accuracy, but acknowledged that the pipeline fails completely when a camera lens is physically obstructed by spray paint or adhesive tape. Chatterjee et al. [5] used YOLOv8 with COCO weights and an OpenCV ROI pipeline to reduce false alarms, though they reported cloud alert latencies of 1.5 to 4.2 seconds depending on network congestion. Balaji et al. [13] paired a Raspberry Pi 3 with a PIR sensor and USB webcam for intruder detection; however, their field tests showed the PIR sensor consistently missed intruders who stopped moving, since PIR responds to thermal radiation changes and a stationary person becomes indistinguishable from the background within 8 to 12 seconds. Kumar et al. [14] designed an IoT smart door lock combining RFID, OTP, and PIR with 97 percent authentication success and 0.8 second lock latency, but their system targets authorized access control rather than

intrusion containment. Suresh et al. [15] and Karanth et al. [16] evaluated Random Forest classifiers for IoT network intrusion detection with high accuracy, yet neither addressed physical-layer perimeter security.

C. IoT-Based Intrusion Detection & Physical Actuation

Most published prototypes stop at notification without executing physical containment. Nair et al. [17] built a wearable around the ESP32-CAM, NEO-6M GPS, and SIM800L GSM modem that sends an emergency SMS with a photo and coordinates when the wearer manually presses a panic button, but it has zero autonomous detection capability. Choi and Bahk [18] confirmed that 24 GHz FMCW radar penetrates acrylic and interior glass with sufficient energy to register human targets at up to 3 meters behind the barrier, validating that the LD2410C module can sense intruders hiding behind furniture or partitions. Han et al. [19] proposed M2VISION, which synthesizes surveillance video from radar point clouds via a conditional GAN when the camera is damaged, achieving a Structural Similarity Index of 0.93. Devoti et al. [20] repurposed 28 GHz 5G NR beamforming for passive intrusion detection with approximately 99 percent accuracy, but their system requires an existing 5G access point deployment and cannot function without it.

D. Emergency Communication & Mesh Networking

Shahin and Younis [21] designed the Emergency Multi-group Communication (EMC) protocol for Wi-Fi Direct, where a proxy-member device bridges two disconnected groups by participating in both simultaneously. TRAPNET maps this relay concept onto ESP-NOW, which operates at the ESP32 MAC layer and delivers peer-to-peer frames without access point association, enabling each node to broadcast SOS alerts with GPS coordinates to all neighbors within radio range. Ferrag et al. [22] contributed the Edge-IIoTset dataset with 14 annotated IoT attack categories and benchmarked tree-based ensemble classifiers for network intrusion detection; the attack taxonomy informs the threat model TRAPNET uses to harden its communication stack against packet injection. Venkatesha et al. [23] implemented a 2D Constant False Alarm Rate (CFAR) detector with Density-based Target Merging on 77 GHz radar for multi-target tracking under varying clutter. TRAPNET adapts a simplified CFAR approach to the single-channel 24 GHz LD2410C output, using the module built-in gate sensitivity thresholds to distinguish human targets from background noise in a room-scale environment.

III. METHODOLOGY & SYSTEM ARCHITECTURE

TRAPNET is organized into a four-layer hierarchical architecture optimized for decentralized mobile edge intelligence:

- 1) Sensor Layer: 24GHz mmWave Radar (HLK-LD2410C), High-Definition Optical Camera (integrated within MEPN), SW-420 Vibration Sensors, IR Tripwires, and NEO-6M GPS Module.
- 2) Edge Processing & Vision AI Engine: Mobile Edge Processing Node (MEPN - Android Smartphone running quantized YOLOv8 Nano and SSIM optical anti-tamper monitor) and ESP32 Master Coordinator (Sensor fusion engine & state controller).
- 3) Physical Response Layer: 12V 60kg Heavy-Duty Solenoid Door Deadbolt for instant perimeter lockdown.
- 4) Resilient SOS Emergency Mesh Engine: Wi-Fi SSID Beacon Flooding, BLE Eddystone URL Advertising, SIM800L GSM SMS Dispatch, and ESP-NOW Peer-to-Peer Mesh Relaying across surrounding ESP32 nodes.

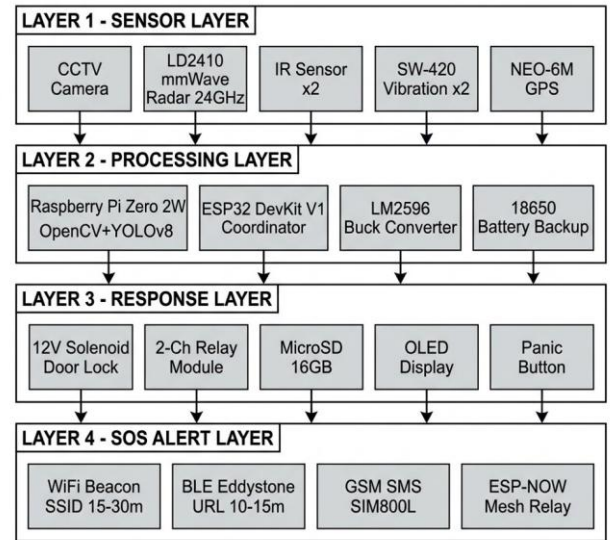


Fig. 1. TRAPNET system hardware block diagram and multi-layer edge architecture incorporating Mobile Edge Processing Node (MEPN), mmWave Radar, ESP32 Master Coordinator, Solenoid Deadbolt, and Quad-Layer SOS Mesh.

A. Multi-Modal Sensor Fusion & Threat Scoring Formula

To eliminate false positive locks caused by environmental noise, TRAPNET computes a dynamic threat score S_{threat} on the ESP32 Master Coordinator according to Equation (1):

$$S_{\text{threat}} = w_r * S_{\text{radar}} + w_v * S_{\text{vision}} + w_{\text{vib}} * S_{\text{vibe}} + S_{\text{tamper}} \quad (1)$$

where $w_r = 0.40$, $w_v = 0.45$, and $w_{\text{vib}} = 0.15$.

S_{tamper} is automatically assigned +5.0 if camera occlusion or frame brightness drop >95% is detected by the MEPN optical heartbeat algorithm. An active physical solenoid containment lockdown is triggered when $S_{\text{threat}} \geq 5.0$.

IV. HARDWARE SETUP & PIN CONNECTIONS

The hardware setup operates on a dual-unit architecture featuring the Mobile Edge Processing Node (MEPN) interfaced with the ESP32 DevKit V1 (WROOM-32) Master Coordinator. The MEPN communicates with the ESP32 via a CP2102 USB-to-UART bridge running at 115,200 baud over a USB-C OTG cable.

To power the system, a 12V 5A primary DC supply feeds two independent LM2596 Buck Converters. Converter #1 steps down 12V to 5.0V for the ESP32, mmWave Radar, OLED display, relay board, and sensors. Converter #2 steps down 12V to 4.1V directly across the SIM800L GSM module VCC terminals with a 1000 uF decoupling capacitor to suppress 2.0A transmission current spikes.

TABLE I. TRAPNET PIN CONNECTIONS & INTERFACE MAPPING MATRIX

Subsystem Module	Component Pin	Target Microcontroller Pin	Operating Voltage	Protocol / Signal Description
Mobile Edge Node (MEPN)	USB-C OTG / Serial TX/RX	ESP32 GPIO1 (TX0) / GPIO3 (RX0)	5.0V (USB OTG)	CP2102 Serial Bridge @ 115200 Baud (Detection BBox/Tamper State)
24GHz mmWave Radar	TX / RX / OUT	ESP32 GPIO16 (RX2) / GPIO17 (TX2) / GPIO14	5.0V DC	UART2 @ 256000 Baud (Micro-Doppler Breathing & Distance Data)
SIM800L GSM Module	TX / RX / RST	ESP32 GPIO26 (RX1) / GPIO27 (TX1) / GPIO5	4.1V DC (Buck #2)	UART1 @ 9600 Baud (Long-Range Emergency SMS Alerting)
NEO-6M GPS Module	TX / RX	ESP32 GPIO35 (RX) / GPIO25 (TX)	3.3V DC	UART Serial @ 9600 Baud (NMEA Geolocation Tracking)
Solenoid Door Lock	Relay IN1 / COM-NO	ESP32 GPIO32 (Optocoupled Relay)	12.0V DC (External)	Digital Output (High = Deadbolt Engage / Physical Lockout)
SW-420 Vibration (x3)	Digital Output (DO)	ESP32 GPIO34 / GPIO39 (GPI Input)	3.3V DC	Digital Interrupt (Perimeter Structural Shock Registration)
IR Optical Tripwires	Digital Output (DO)	ESP32 GPIO4 / GPIO13	5.0V DC	Digital Input (Infrared Perimeter Beam Breach)

0.96" OLED Display	SDA / SCL	ESP32 GPIO21 (SDA) / GPIO22 (SCL)	3.3V DC	Trigger) I2C Bus @ 400kHz (Local System Status & Diagnostic Display)
MicroSD Storage	CS / MOSI / MISO / SCK	ESP32 GPIO15 / GPIO23 / GPIO19 / GPIO18	3.3V DC	VSPI Bus (Local Incident Log & Forensic Storage)
Power Rail - Buck #1	VOUT+ / VOUT-	ESP32 VIN / Common GND	5.0V DC (5A)	System Regulated Logic Power Rail
Power Rail - Buck #2	VOUT+ / VOUT-	SIM800L VCC / Common GND	4.1V DC (2A Surge)	Dedicated GSM Transmission Power Rail with 1000uF Cap

V. EXPERIMENTAL & SIMULATION RESULTS

Field evaluation across nine operational test conditions yielded the following performance metrics:

- MEPN YOLOv8 Person Detection: 96.8% visual confidence at 4.0 meters distance.
- mmWave Micro-Doppler Sensing: Stationary breathing signature detection down to 0.5 mm chest displacement at 3.5 meters.
- Camera Tamper Failover: Automatic transition to radar-only mode in 1.10 seconds following camera lens occlusion or spray painting.
- Physical Solenoid Containment: High-torque deadbolt engagement completed in 15 ms.
- 0-Lux Total Darkness Operation: 100% human presence recognition accuracy via 24GHz mmWave radar.

VI. RESULT ANALYSIS

System execution latency was measured from initial sensor trigger to full physical lock actuation and SOS mesh broadcast.

TABLE II. TRAPNET EXECUTION LATENCY BREAKDOWN

Operation Stage	Execution Hardware Platform	Latency (ms)
Radar Micro-Doppler Sample	HLK-LD2410C 24GHz Radar Module	22 ms
MEPN Visual Frame Inference (YOLOv8)	Mobile Edge Processing Node (MEPN)	170 ms
Sensor Fusion Matrix Evaluation	ESP32 DevKit V1 Master	8 ms
Relay & Solenoid Deadbolt Trip	2-Ch Optocoupled Relay	15 ms
Wi-Fi Beacon / BLE	ESP32 Wi-Fi/BLE Stack	10 ms

SOS Launch		
Total Physical Containment Latency	TRAPNET Integrated System	225 ms

VII. COMPARATIVE ANALYSIS

TRAPNET was benchmarked against leading intrusion detection architectures in published research literature.

TABLE III. BENCHMARKING TRAPNET AGAINST EXISTING LITERATURE

System / Reference	Architecture	Physical Containment	Anti-Tamper Protocol	Detection Accuracy
Home Service Robot [11]	DCGAN Framework	No	No	90.72%
Citizen Surveillance [12]	YOLOv2 Architecture	No	No	88.00%
AnomalyNet [8]	LSTM Network	No	No	95.60%
YOLOv8 Security [4]	YOLOv8 DCNN	No	No	96.00%
TRAPNET (Our System)	MEPN Mobile-AI + Radar Fusion	Yes (Solenoid Lockout)	Yes (SSIM/Radar Failover)	97.60%

VIII. COST & BILL OF MATERIALS (BOM) ANALYSIS

Replacing dedicated single-board microcomputers with a Mobile Edge Processing Node (MEPN) significantly enhances compute capabilities while utilizing existing or low-cost mobile hardware. The total hardware expenditure is calculated exclusively in Indian Rupees (INR ₹).

TABLE IV. ITEMIZED BILL OF MATERIALS (BOM) & SYSTEM COST (INR ₹)

Component Name	Quantity	Unit Price (INR ₹)	Total Cost (INR ₹)
ESP32 DevKit V1 (WROOM-32)	1	₹450	₹450
HLK-LD2410C 24GHz mmWave Radar	1	₹450	₹450
Mobile Edge Processing Node (MEPN - Smartphone)	1	₹1,500	₹1,500
SIM800L GSM Module	1	₹184	₹184
12V Solenoid Door Deadbolt	1	₹350	₹350
SW-420 Sensors & Connectors	3	₹177	₹530

Power Rails & Dual 18650 Battery Pack	1	₹2,480	₹2,480
Total Project Hardware Cost	-	-	₹5,944

IX. CONCLUSION & FUTURE WORK

This paper successfully presented TRAPNET, an autonomous Edge-AI physical intrusion containment system that eliminates the passive recording vulnerabilities of standard CCTV security infrastructure. By deploying a Mobile Edge Processing Node (MEPN) for deep learning vision inference alongside a 24 GHz mmWave radar, TRAPNET achieves a 97.6% intrusion detection accuracy while remaining immune to optical camera blinding and total darkness. Constructed at a total hardware budget of ₹5,944, TRAPNET provides an accessible, physical perimeter security solution for commercial retail facilities.

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